FINM3013 Introduction to Financial Derivatives

Chapter 9 Exercises

1. A trader has a put option contract to sell 100 shares of a stock for a strike price of $60. What is the effect on the terms of the contract of:
   (a) A $5 dividend being declared
   (b) A $5 dividend being paid
   (c) A 5-for-2 stock split
   (d) A 5% stock dividend being paid.

2. A trader writes five naked put option contracts, with each contract being on 100 shares. The option price is $10, the time to maturity is six months, and the strike price is $64.
   (a) What is the margin requirement if the stock price is $58?
   (b) How would the answer to (a) change if the rules for index options applied?
   (c) How would the answer to (a) change if the stock price were $70?
   (d) How would the answer to (a) change if the trader is buying instead of selling the options?

3. The price of a stock is $40. The price of a one-year European put option on the stock with a strike price of $30 is quoted as $7 and the price of a one-year European call option on the stock with a strike price of $50 is quoted as $5. Suppose that an investor buys 100 shares, shorts 100 call options, and buys 100 put options. Draw a diagram illustrating how the investor's profit or loss varies with the stock price over the next year. How does your answer change if the investor buys 100 shares, shorts 200 call options, and buys 200 put options?